

National College of Ireland

Project Submission Sheet

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Programme: MSc in Cloud Computing **Year:** 2026 - 2027

Module: Research in Computing

Lecturer: Bernardo Pulido-Gaytan

Submission Due Date: 12 June 2026

Project Title: FinOps in Cloud Computing:
Evaluating Cost Governance for Reducing
Unnecessary Cloud Spending

Word Count: 739

Academic Integrity Certificate Verification Code: ZQJtpeqV2a

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Signature: Pooja

Date: 12 June 2026

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Research In Computing

FinOps in Cloud Computing: Evaluating Cost Governance for Reducing Unnecessary Cloud Spending

Your Name/Student Number	Course	Date
Pooja	MSc in Cloud Computing	12/06/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

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[Insert Tool Name]	
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[Insert Sample prompt]	[Insert Sample response]

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Additional Evidence:

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FinOps in Cloud Computing: Evaluating Cost Governance for Reducing Unnecessary Cloud Spending

Pooja

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MSc in Cloud Computing

12th June 2026

Abstract

Cloud computing offers scalability and agility with services however, cloud cost management continues to be an issue organisations globally struggle with. Often organisations overspend budgets because resources that are not tagged or used are left unchecked. FinOps allows companies to better cloud cost management by having finance, operations and engineering collaborate. This research aims to explore cloud cost governance and proposes a Cost Governance Effectiveness (CGE) measure to quantify the value added by tagging policies and automated controls to minimise avoidable cloud spend. Proposed research will be low-cost and reproducible for organisations to better their cloud financial management practices. This research is especially beneficial for those with small to medium-sized organisations.

1 Introduction

Cloud platforms are used by organisations to store data, execute applications, and scale services according to demand. While cloud platforms allow organisations to be flexible and manage their resources more efficiently, cloud cost optimisation has been one of the key challenges faced by organisations on cloud platforms. Resources are often left idle or unmonitored on cloud platforms, leading to avoidable cloud spend.

This research paper is centred around cloud cost governance. It analyses how unnecessary cloud spend can be minimised by implementing proper resource tagging and automated cost control policies. It also proposes a Cost Governance Effectiveness (CGE) score which is an easy way to measure how well cloud resources are governed and the amount of unnecessary spend that can be eliminated.

2 Motivation

A business driver for this research initiative is based on the gap between expected vs actual spend across many organizations. Businesses are often spending money for cloud resources they don't use, oversized, untagged, and poorly monitored. This behavior drives unwanted cloud costs and lowers monetary effectiveness.

Cloud Cost Governance will be explored throughout this research project. There will be a study focused on reducing unnecessary cloud spend through proper resource tagging, and automated policy restrictions. A Cost Governance Effectiveness (CGE) will be created to help benchmark governance effectiveness.

FinOps practices have been a major topic in recent industry articles and research papers showing their prominence in providing cloud visibility and accountability. There have been limited studies that show-case how to measure reductions to unnecessary cloud spend derived from governance policy.

The practicality of this project stems from cloud providers like AWS and Azure offering low-cost utilities for tagging, budgets, monitoring, and cost analysis.

3 Research Question

The driving research question:

Can unnecessary cloud spend be reduced through improved cloud resource tagging and policy based controls as part of cloud cost governance and how well can this be measured through use of a Cost Governance Effectiveness (CGE) measure?

We will be measuring visibility of cost and avoidable spend in cloud environments and what factors influence your ability to govern your cloud costs.

4 Objectives

The following objectives will allow us to answer our research question:

1. Provision a cloud environment to test various cloud cost governance best practices.
2. Identify idle cloud resources and assess visibility into cloud costs.
3. Implement tagging and policy-based cost controls (budgets/alerts)
4. Create a Cost Governance Effectiveness (CGE) metric to benchmark cloud cost management
5. Offer suggestions to eliminate wasteful cloud spending.

Beneficiaries

Cloud administrators, FinOps professionals, and finance teams who need visibility into and control over cloud costs stand to benefit the most from this research. Additionally, because the approach we propose is low cost and straightforward to implement in cloud environments, small and medium sized organisations can also take advantage of our work.

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